Embedded Control for Robotics

Lecture # 16 Introduction To Filters

Book: Embedded Digital Control with Microcontrollers

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Overview Today's Lecture

- Filters
- Types of Filters
- Transfer Function of filters
- Circuits of Filters

Filters

Filter is a circuit that modifies a signal to remove noise and other unwanted frequencies, allowing the system to perform more smoothly and accurately.

Filters select certain frequencies to pass while rejecting or attenuating others, and they are crucial for improving signal quality and preventing issues like unstable control actions or mechanical wear.

Filters may be classified as either digital or analog.

- *Digital filters* are implemented using a digital computer or special purpose digital hardware.
- *Analog filters* may be classified as either passive or active and are usually implemented with R, L, and C components and operational amplifiers.

Filters

- An *active filter* is one that, along with R, L, and C components, also contains an energy source, such as that derived from an operational amplifier.

- A *passive filter* is one that contains only R, L, andC components. It is not necessary that all three be present. L is often omitted (on purpose) from passive filter design because of the size and cost of inductors – and they also carry along an R that must be included in the design.

Application in Control Systems

Examples of applications

- **Motor speed control:** Filtering a motor's speed sensor signal prevents rapid, fluctuating control actions caused by measurement noise, ensuring a stable and precise speed.
- **Sensor interfaces:** Digital or analog filters can be used to smooth noisy sensor readings before they are used in the control loop.
- **Power electronics:** Filters are used to suppress voltage spikes and reduce electromagnetic interference (EMI).

Filters: Background:

- Generally speaking, digital filters have become the focus of attention in the last 40 years. The interest in digital filters started with the advent of the digital computer, especially the affordable PC and special purpose signal processing boards. People who led the way in the work (the analysis part) were Kaiser, Gold and Radar.
- A digital filter is simply the implementation of an equation(s) in computer software. There are no R, L, C components as such. However, digital filters can also be built directly into special purpose computers in hardware form. But the execution is still in software.

Passive Analog Filters

Four types of filters - "Ideal"

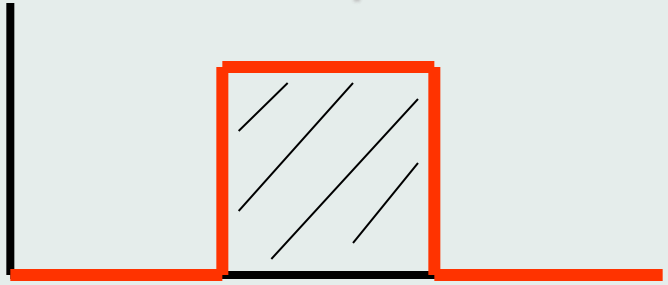
lowpass



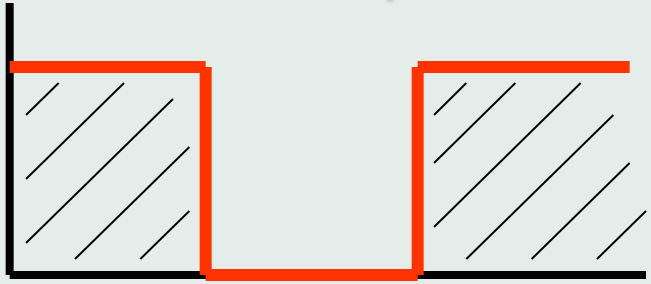
highpass



bandpass



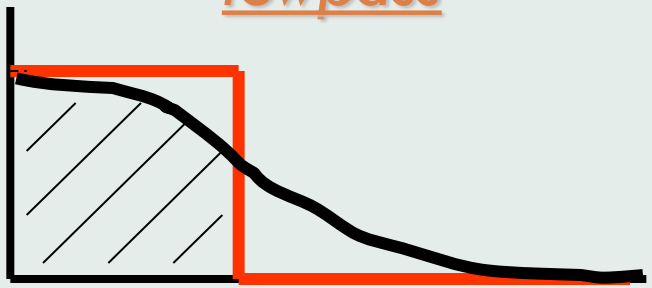
bandstop



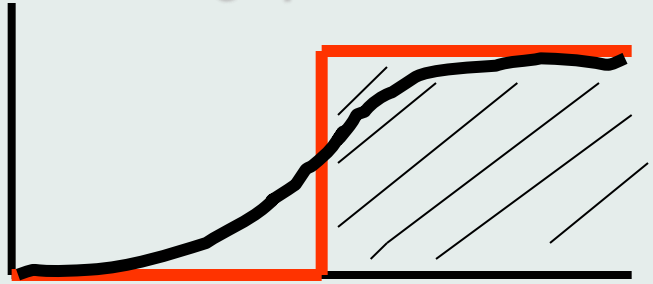
Passive Analog Filters

Realistic Filters:

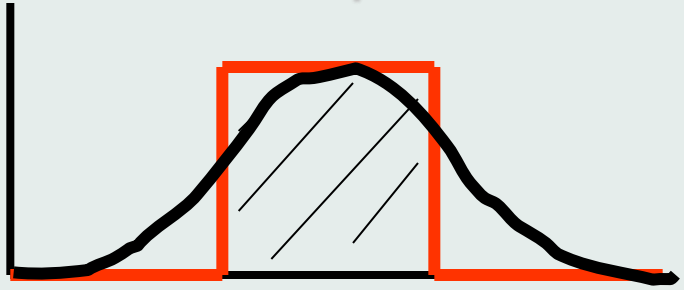
lowpass



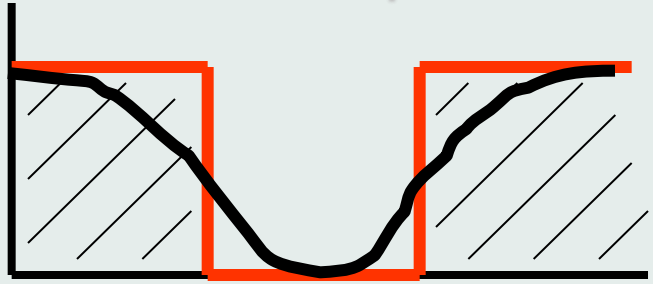
highpass



bandpass



bandstop

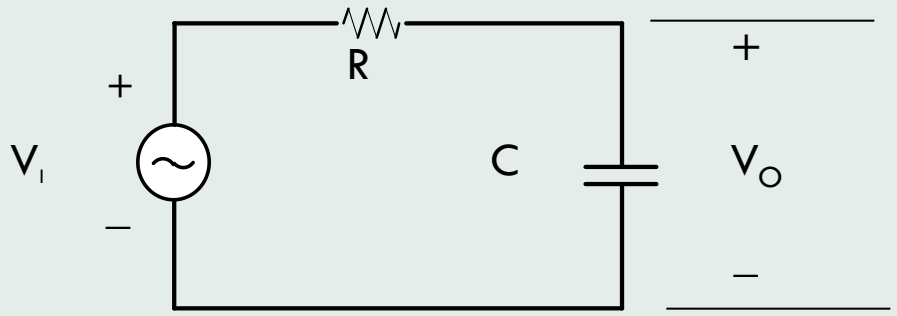


Passive Analog Filters

Kalman filters: A more advanced type of filter used in control systems and robotics to estimate the state of a process based on noisy measurements.

Low Pass Filter

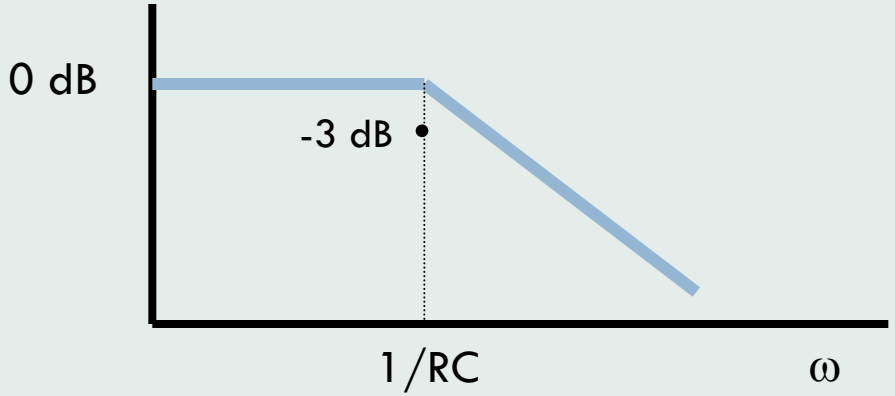
Consider the circuit below.



Low pass filter circuit

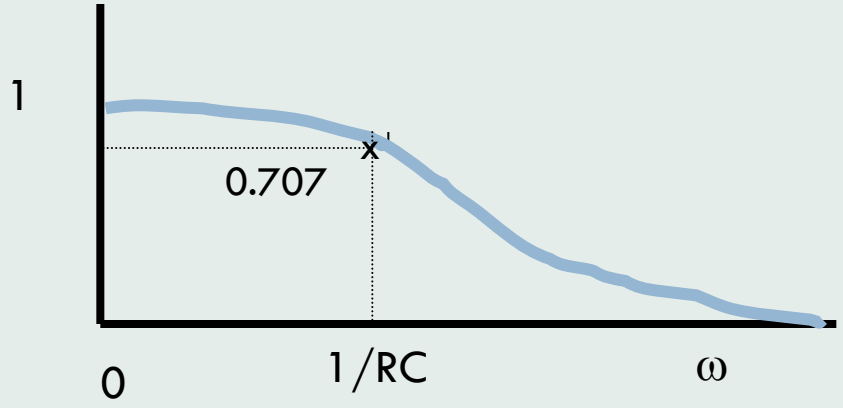
$$\frac{V_o(jw)}{V_i(jw)} = \frac{\frac{1}{jwC}}{R+\frac{1}{jwC}} = \frac{1}{1+jwRC}$$

Low Pass Filter



Bode

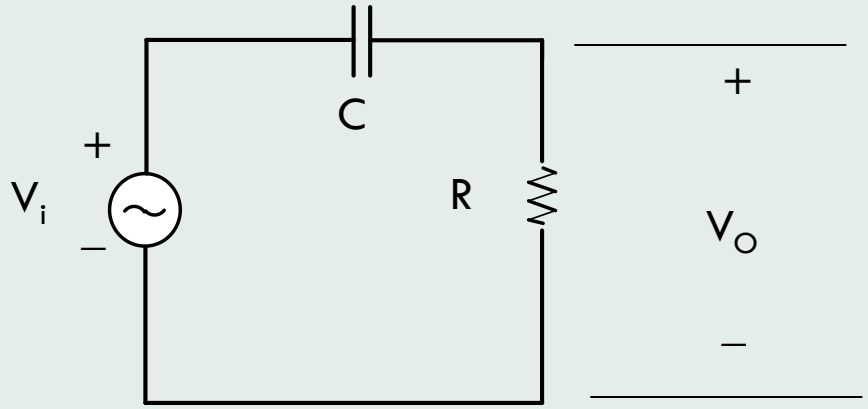
Passes low frequencies
Attenuates high frequencies



Linear Plot

High Pass Filter

Consider the circuit below.

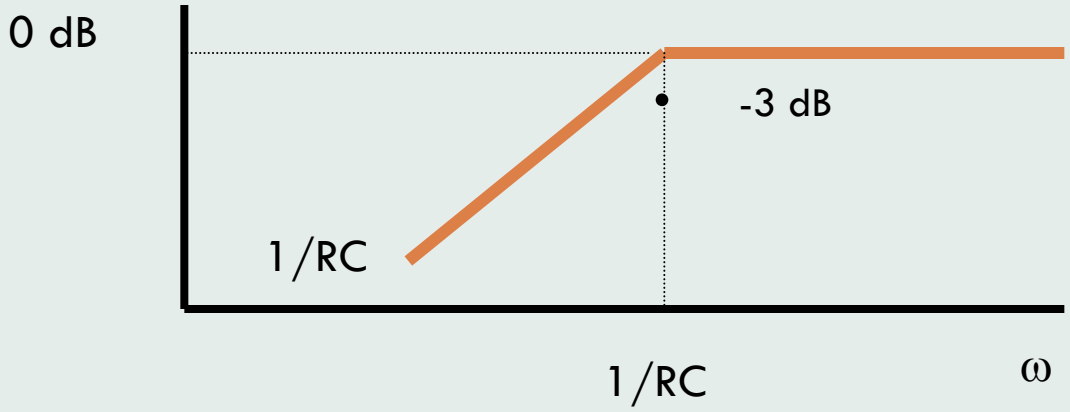


High Pass Filter

$$\frac{V_o(jw)}{V_i(jw)} = \frac{R}{R+\frac{1}{jwC}} = \frac{jwRC}{1+jwRC}$$

High Pass Filter

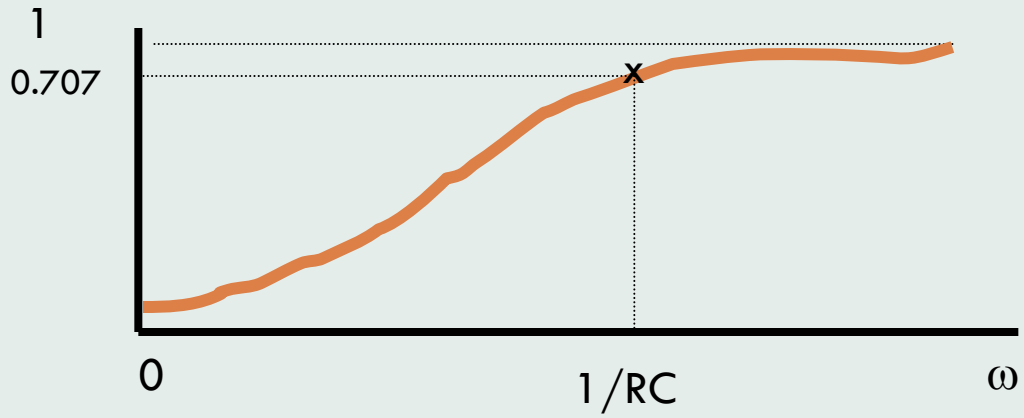
Bode



Passes high frequencies

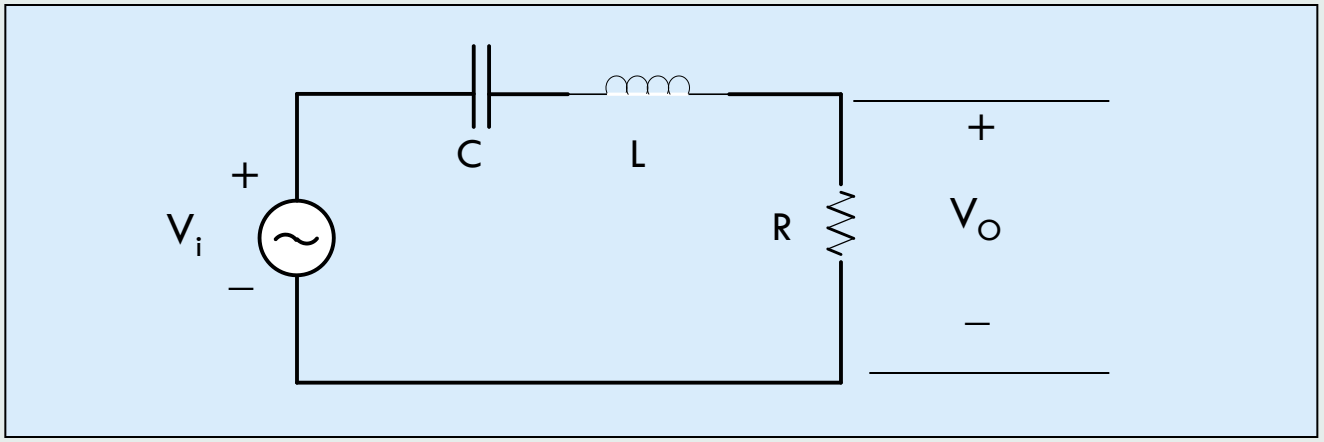
Attenuates low frequencies

Linear



Bandpass Pass Filter

Consider the circuit shown below:

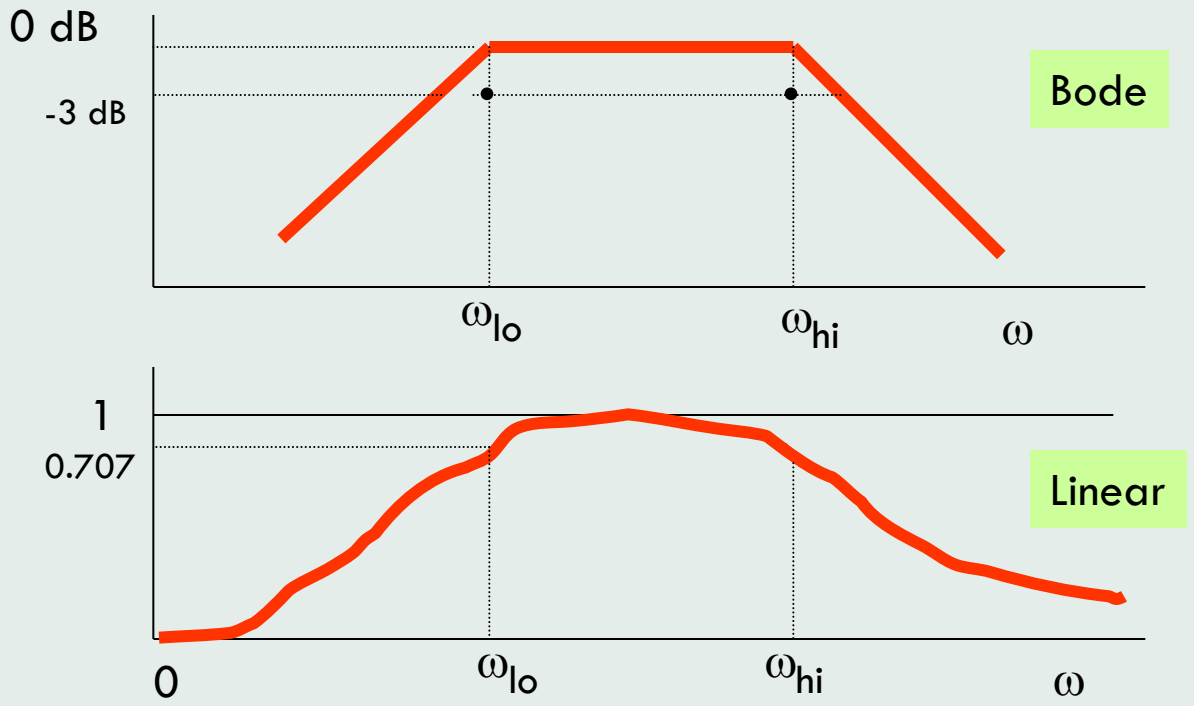


When studying series resonant circuit we showed that;

$$\frac{V_O(s)}{V_i(s)} = \frac{\frac{R}{L}s}{s^2 + \frac{R}{L}s + \frac{1}{LC}}$$

Bandpass Pass Filter

We can make a bandpass from the previous equation and select the poles where we like. In a typical case we have the following shapes.



Bandpass Pass Filter: Example

Suppose we use the previous series RLC circuit with output across R to design a bandpass filter. We will place poles at –200 rad/sec and – 2000 rad/sec hoping that our –3 dB points will be located there and hence have a bandwidth of 1800 rad/sec. To match the RLC circuit form we use:

$$\frac{2200s}{s^2+2200s+400000} = \frac{2200s}{(s+200)(s+2000)} = \frac{2200s}{200x2000(1+\frac{s}{200})(1+\frac{s}{2000})}$$

The last term on the right can be finally put in Bode form as;

$$\frac{0.0055\,jw}{(1+\frac{jw}{200})(1+\frac{jw}{2000})}$$

Bandpass Pass Filter: Example

From this last expression we notice from the part involving the zero we have in dB form;

$$20\log(.0055) + 20\log w$$

Evaluating at w = 200, the first pole break, we get a 0.828 dB what this means is that our –3dB point will not be at 200 because we do not have 0 dB at 200. If we could lower the gain by 0.829 dB we would have – 3dB at 200 but with the RLC circuit we are stuck with what we have. What this means is that the – 3 dB point will be at a lower frequency. We can calculate this from

$$\log\frac{200}{wlow}\,x20\,\frac{dB}{dec} = 0.828\,dB$$

Bandpass Pass Filter: Example

This gives an $w_{low} = 182$ rad/sec. A similar thing occurs at w_{hi} where the new calculated value for w_{hi} becomes 2200. These calculations do no take into account a 0.1 dB that one pole induces on the other pole. This will make w_{lo} somewhat lower and w_{hi} somewhat higher.

One other thing that should have given us a hint that our w_1 and w_2 were not going to be correct is the following:

$$\frac{\frac{R}{L}s}{(s^2 + \frac{R}{L}s + \frac{1}{LC})} = \frac{(w_1 + w_2)s}{(s^2 + (w_1 + w_2)s + w_1 w_2)}$$

What is the problem with this?

Bandpass Pass Filter: Example

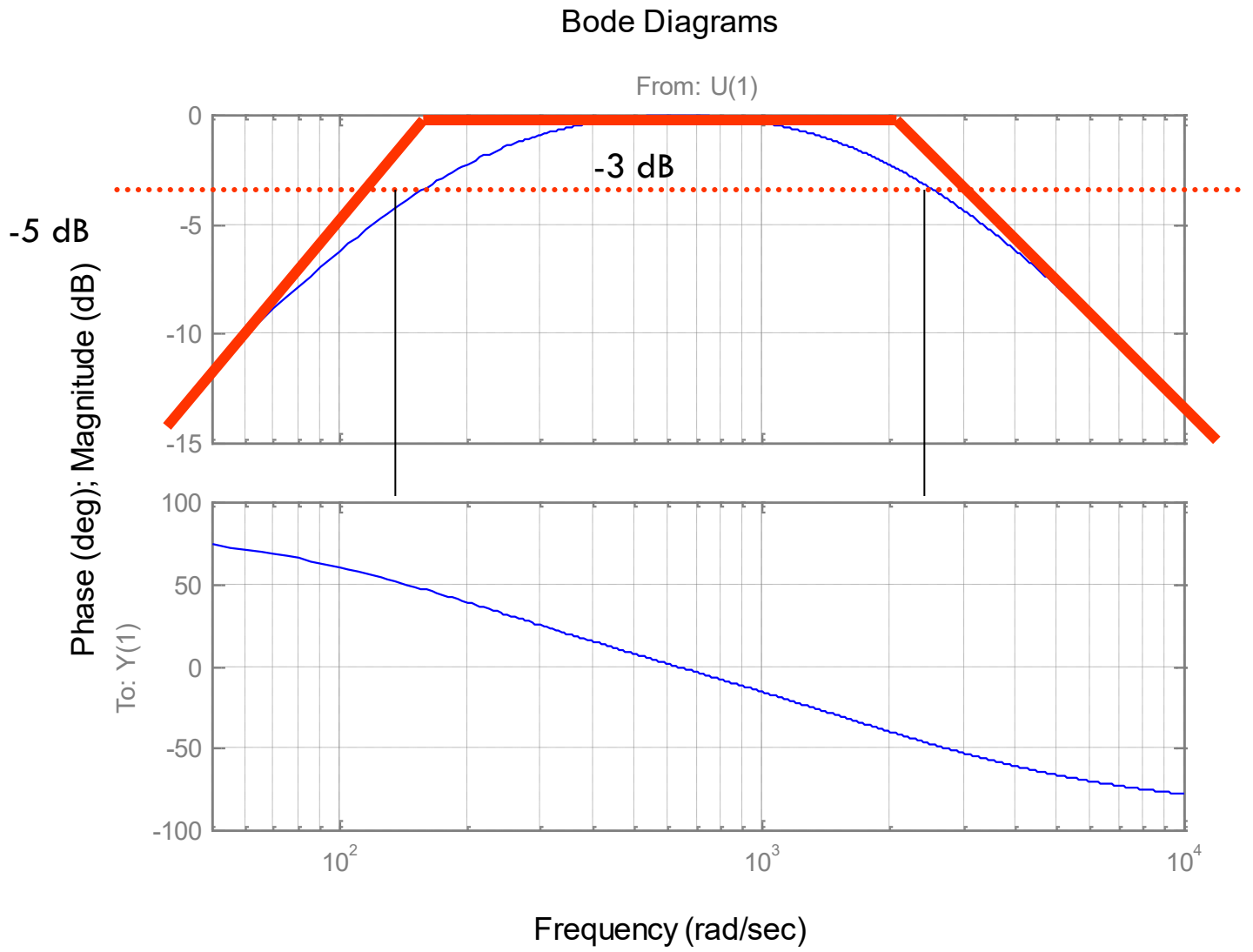
The problem is that we have

$$\frac{R}{L} = (w_1 + w_2) = BW = (w_2 - w_1)$$

Therein lies the problem. Obviously the above cannot be true and that is why we have aproblem at the –3 dB points.

We can write a Matlab program and actually check all of this.
We will expect that w_1 will be lower than 200 rad/sec and w_2 will be higher than 2000 rad/sec.

Bandpass Pass Filter: Example



Bandpass Pass Filter: Example

Perhaps going in the direction to stimulate your interest in taking a course on filtering, a 10 order analog bandpass butterworth filter will be simulated using Matlab. The program is given below.

```
N = 10;      %10th order butterworth analog prototype

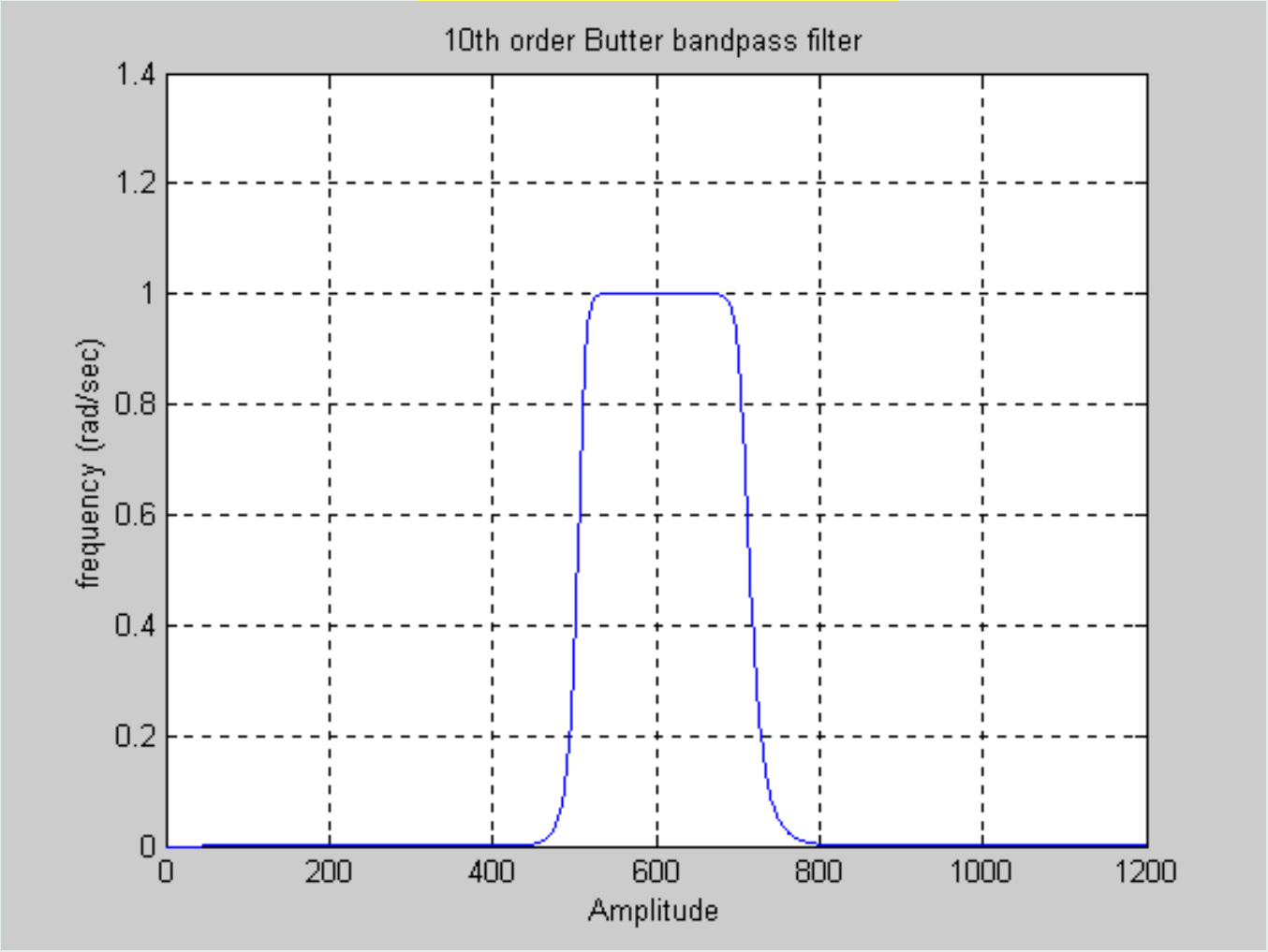
[ZB, PB, KB] = buttap(N);
numzb = poly([ZB]);
denpb = poly([PB]);

wo = 600;  bw = 200;       % wo is the center freq
                    % bw is the bandwidth
[numbbs,denbbs] = lp2bs(numzb,denpb,wo,bw);

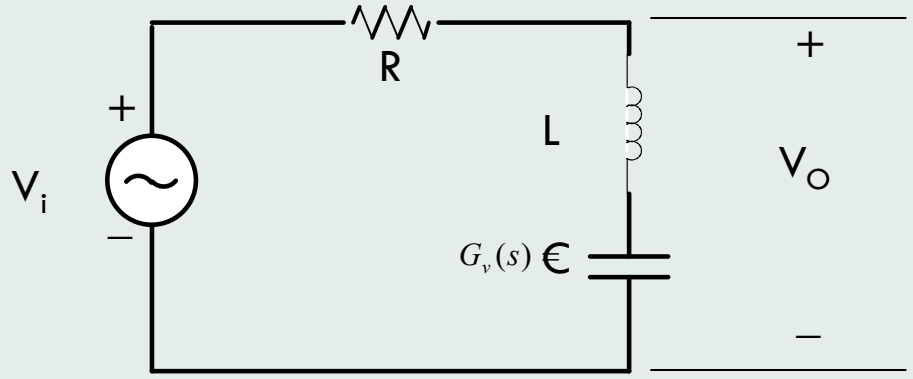
w = 1:1:1200;

Hbbs = freqs(numbbs,denbbs,w);
Hb = abs(Hbbs);
plot(w,Hb)
grid
xlabel('Amplitude')
ylabel('frequency (rad/sec)')
title('10th order Butterworth filter')
```

Bandpass Pass Filter: Example



RLC Band stop Filter



The transfer function for V_O/V_i can be expressed as follows:

$$G_v(s) = \frac{s^2 + \frac{1}{LC}}{s^2 + \frac{R}{L}s + \frac{1}{LC}}$$

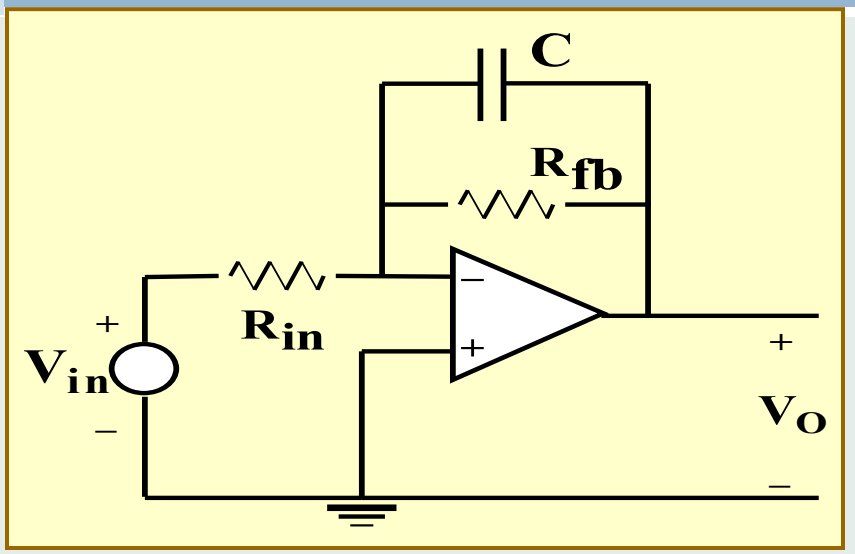
RLC Band Stop Filter

This is of the form of a band stop filter. We see we have complex zeros on the jw axis located

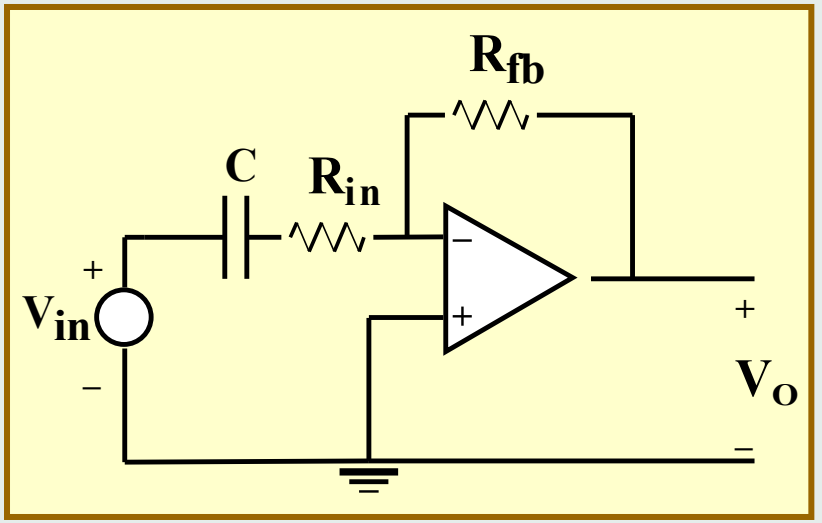
$$\pm j \frac{1}{\sqrt{LC}}$$

From the characteristic equation we see we have two poles. The poles an essentially be placed anywhere in the left half of the s-plane. We see that they will be to the left of the zeros on the jw axis.

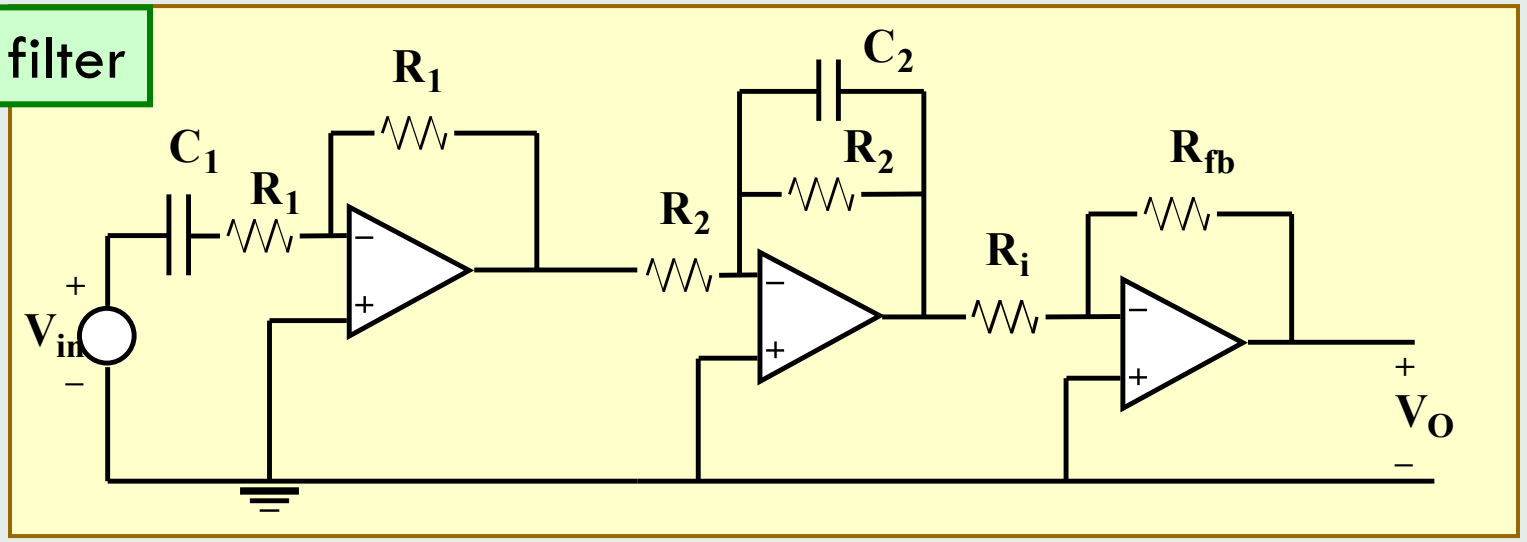
Circuit of Low pass filter



Circuit of High pass



Band pass filter



Circuit of Band stop filter

